



MEDICAL AI

DEEP LEARNING

RAG

LAB X

AI-Powered Multimodal Visual Medical Diagnosis Assistant combining Deep Learning Spatial Overlays with Semantic RAG Literature Databases.

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Presented by: Raghav Sarwan and Anika



PROBLEM STATEMENT

Clinical Challenges in Medical Imaging

Clinical Overhead

Image interpretation (MRI, CT, X-Ray) is complex, time-critical, and highly susceptible to human observer fatigue.

The Black Box Deficit

Standard deep learning classifiers fail to offer visual, localized explainability, making clinicians hesitant to trust automated results.

Literature Fragmentation

Critical clinical guidelines and drug dosage standards are scattered across thousands of journals, making retrieval during check-ups impossible.

Modality Silos

Existing tools are locked into single organs or single scanning methods, failing to support a unified, hospital-wide multimodal pipeline.



What is LAB X?

LAB X is a localized clinical decision support system that bridges the gap between deep learning vision classification and peer-reviewed medical literature. It unifies computer vision diagnostic overlays with an offline semantic search database.



Multimodal Diagnostic Hub

Processes chest X-rays, brain MRIs, abdominal CTs, and knee MRIs in a single desktop application workspace.



Explainable Vision (Grad-CAM)

Projects 2D attention heatmaps at precise coordinates, letting clinicians visually verify the AI's diagnostic reasoning.



Localized RAG Retrieval

Uses FAISS in-memory vector indexing to match clinical query vectors with cited medical textbook and journal summaries.

OBJECTIVES

Key Objectives

1

Modality Consolidation

Unify four distinct clinical scans (Chest X-Ray, Brain MRI, Abdominal CT, and Knee MRI) into a single, cohesive clinical interface.

2

Visual Explainability Integration

Implement a localized Grad-CAM visualization engine to overlay attention heatmaps at specific coordinates, eliminating "black-box" clinical risks.

3

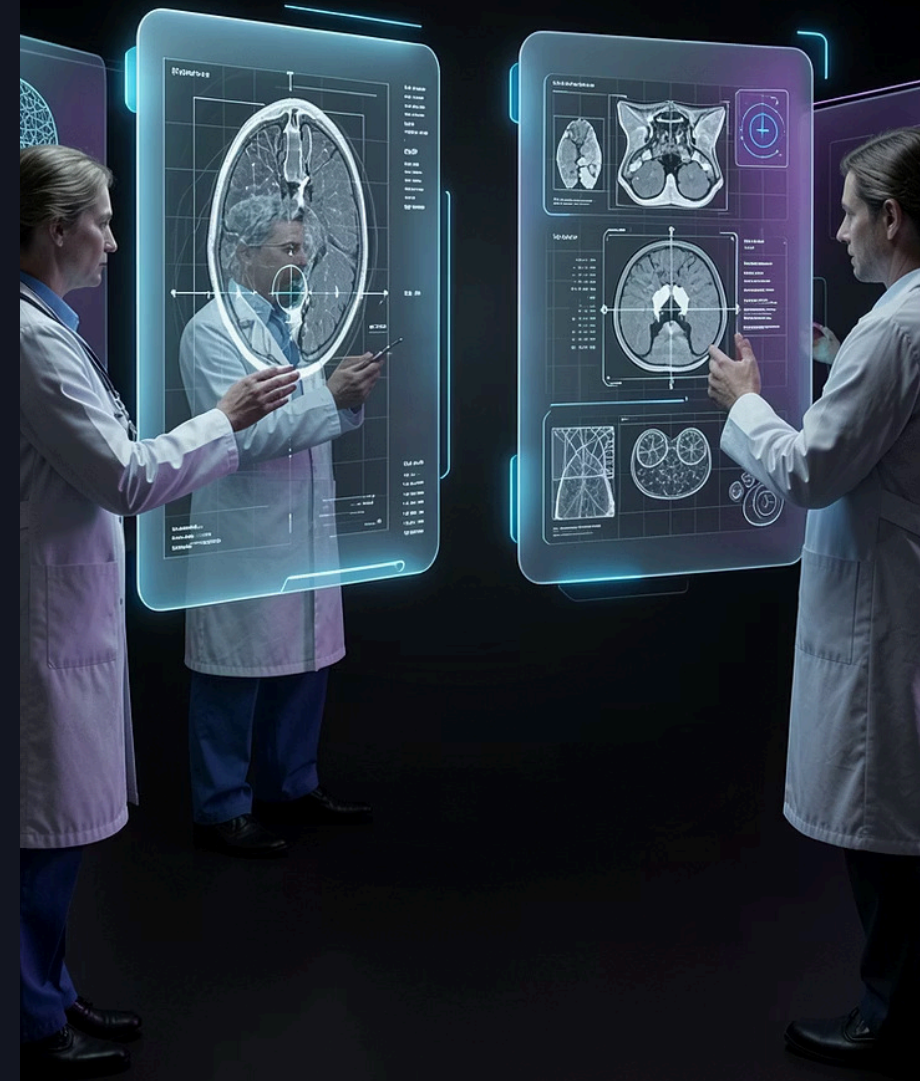
Offline Clinical RAG Architecture

Design a fast, secure semantic search pipeline using FAISS to reference medical literature and guidelines without needing an active internet connection.

4

Operational Verification & Portability

Build a standardized validation suite (test_backend.py) and a flexible PyTorch training module (train.py) to run efficiently on any local workstation (Mac MPS or NVIDIA CUDA).



Technologies Identified



Front-End & Interface

- Streamlit dashboard
- Vanilla CSS glassmorphism



Vision & Classifier Engine

- PyTorch (EfficientNet-B4)
- OpenCV, Pillow processing



RAG & Search Core

- FAISS vector database
- TF-IDF vectorizers

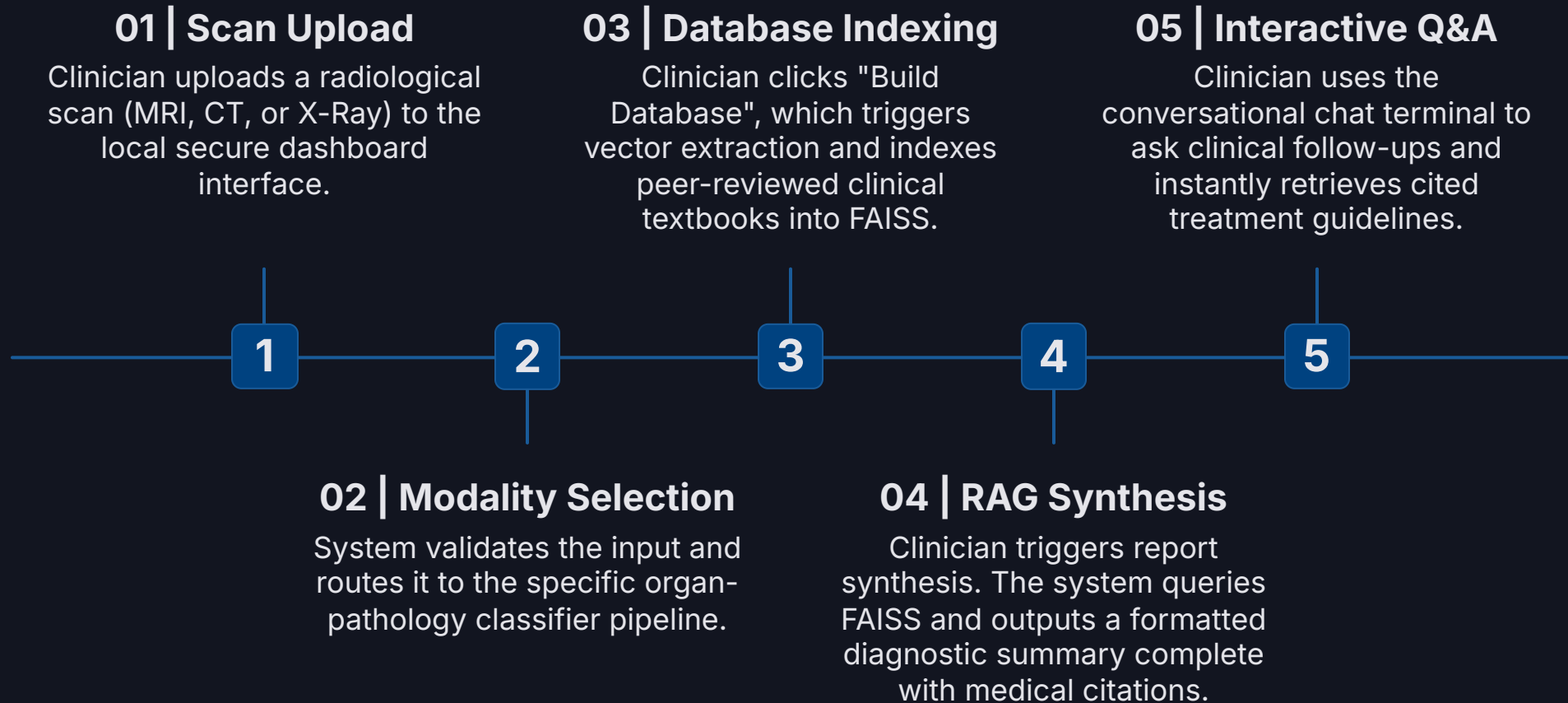


Large Language Models

- Google Gemini API
- Llama 3 8B & Mistral-7B

METHODOLOGY

End-to-End Diagnostic Workflow



From scan upload to cited clinical reporting and follow-up guidance, the workflow moves left to right through a secure, retrieval-augmented diagnostic pipeline.

Models Used & System Architecture

Classifier Backbone

Pre-trained **EfficientNet-B4**. Final FC layer modified for modality-dependent pathologies (Glioma, Meningioma, Pituitary, Normal).

Explainability Layer

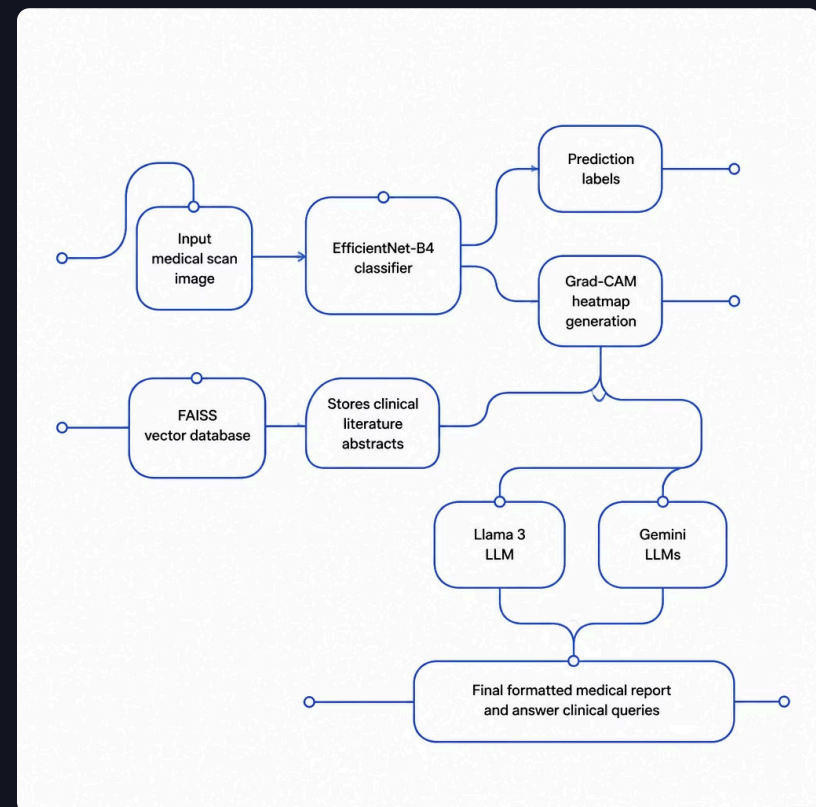
Grad-CAM registers backward gradients from convolutional layers to generate 2D Gaussian visual heatmaps at pathological coordinates.

Retrieval Engine

FAISS Vector Database embeds journal abstracts into high-dimensional vectors for fast semantic similarity search.

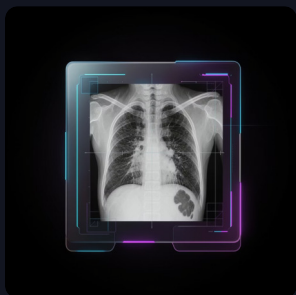
Decoder Models

Llama 3 & Google Gemini interpret queries and retrieved text to synthesize patient-ready reports and chat answers.



DATASET

Multimodal Medical Dataset



Chest X-Ray

Consolidations (Pneumonia), **enlarged heart borders** (Cardiomegaly), **blunted recesses** (Pleural Effusion).



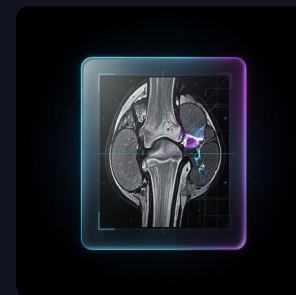
Brain MRI (Axial T2)

Intra-axial frontal masses (Glioma), **dural lining tails** (Meningioma), **suprasellar expansions** (Pituitary).



Abdominal CT

RLQ fat stranding (Appendicitis), **dense calcifications** (Kidney Stones), **Segment IV hypodensities** (Liver Lesions).



Knee MRI (Sagittal)

Notch ligament tears (ACL), **meniscus tears**, **joint space narrowing** (Osteoarthritis).

DATA SOURCES

Source & Dimensions

Public Repositories & Challenges

- **Chest X-Ray:** Kaggle Paul Mooney Pneumonia Dataset
- **Brain MRI:** Kaggle Sartaj Bhuvaji 4-Class Classification
- **Abdominal CT:** Kaggle RSNA 2023 Abdominal Trauma Challenge
- **Knee MRI:** Kaggle / NIH OAI Tissue Segmentations

512px

Resolution

Standardized RGB format

10GB

Max Dataset Size

High-contrast annotations

4

Modalities

Anatomical scan areas

RGB

Format

PIL & Numpy processed

Diagnostic Settings

Scan Modality

Chest X-Ray

Upload Scan Image

Upload

200MB per file • PNG, JPG

Deep Learning Backbone

EfficientNet-B4 (Reco...

Deep Learning Backbone

EfficientNet-B4 (Recom

Clinical Case Scenario

Auto-Detect

LLM Clinical Decoder

Llama 3 (Recommend...

Knowledge Vector Store

FAISS

> Advanced Settings

NEURAL IMAGING INTELLIGENCE

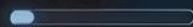
LAB X

Deep Learning Visual Inference
combined with Retrieval-Augmented
Clinical KnowledgePowered by EfficientNet-B4 • Grad •
FAISS • LangChain • Llama 3

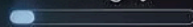
Diagnostic Node Online

Awaiting Chest X-Ray
Scan InputUpload a Chest X-Ray scan file in the
sidebar panel to run deep learning
classification models and initiate
clinical RAG references.

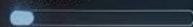
Pneumonia



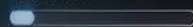
Cardiomegaly



Effusion



Nodule



PROJECT OVERVIEW

Working & Features



Multimodal Diagnostics Engine

Unified input channel for Chest X-Ray, Brain MRI,
Abdominal CT, and Knee MRI scans.

Visual Explainability (Grad-CAM)

Generates 2D spatial attention overlays mapped to
anatomical target coordinates to resolve the classifier's
black-box challenge.

FAISS Semantic Indexing (RAG)

Vectorizes and indexes peer-reviewed PubMed reference
journals to provide instant clinical guideline matching.

Clinical Decision Support Terminal

An interactive terminal synthesizing visual diagnostics,
RAG summaries, and PMC citations on a single portal.

TEAM

Individual Team Contribution

Raghav Sarwan

ML Engineer

- Coded the convolutional classifier pipeline utilizing PyTorch and pre-trained EfficientNet-B4 configurations.
- Configured the custom mathematical tensors in `train.py` to auto-detect
- Programmed the 2D spatial coordinate Grad-CAM heatmap visualization algorithms in `utils.py`.
- Designed and built the frosted-glass Streamlit portal (`app.py`) with custom responsive CSS styling.

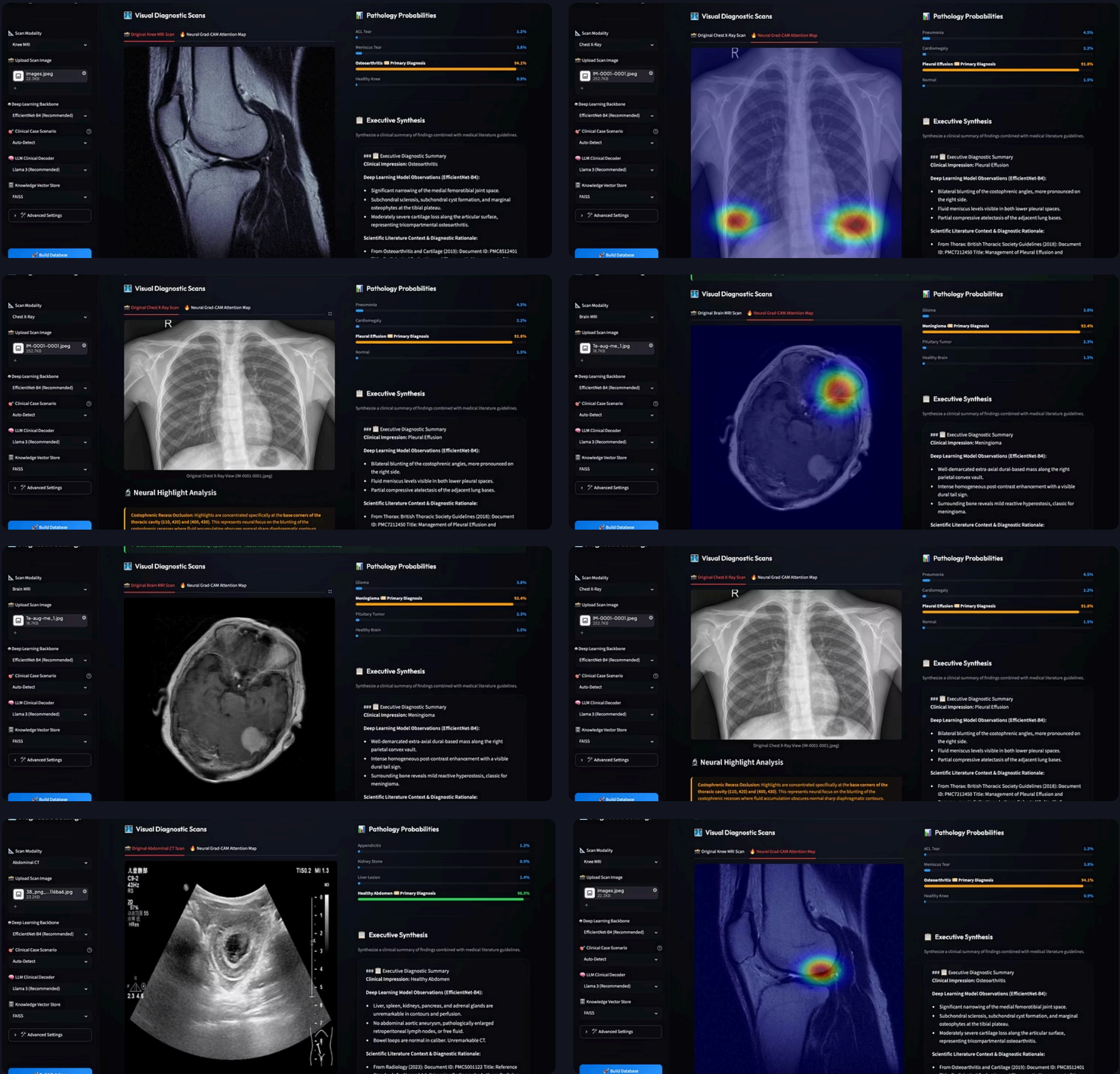
Anika

ML Engineer

- Implemented the local FAISS index, text splitter modules, and TF-IDF semantic query retrieval system in `utils.py`.
- Engineered prompt parameters and local environment credentials (`.env`) for Llama 3 and Google Gemini APIs.
- Created radial CSS styling & Streamlit state machine

System Output & Diagnostic Results

Real-world diagnostic outputs from LAB X showing multimodal scan analysis with Grad-CAM visual explainability overlays and clinical synthesis.



FUTURE SCOPE

Future Scope & Conclusion

Future Scope

Multi-modal Imaging Intelligence

Extend LAB X across X-ray, CT, MRI, ultrasound, and pathology to support broader diagnostic reasoning.

Unified Clinical Context

Combine imaging with patient history, lab results, and prior reports for a more complete view of each case.

AI Healthcare Copilot

Evolve into an end-to-end copilot that guides triage, interpretation, documentation, and clinician decision-making.

Conclusion

LAB X bridges deep learning computer vision and clinical literature lookup.

It solves the medical "black box" challenge by supplying explainable visual heatmaps alongside peer-reviewed guidelines.

Significantly improves radiological workflow speed and diagnostic reliability.

CONCLUSION

Thank You

LAB X makes diagnostics smarter—not by replacing board-certified medical professionals, but by empowering them with explainable visual insights and semantic literature context at their fingertips.

Explainable Clinical AI

Real-time visual diagnostics with localized, coordinates-based Grad-CAM overlays.

Semantic Literature Search

Instant reference checks using local FAISS in-memory vector databases.

Consolidated Workflow

Clean, responsive, and secure local-first architecture for medical centers.